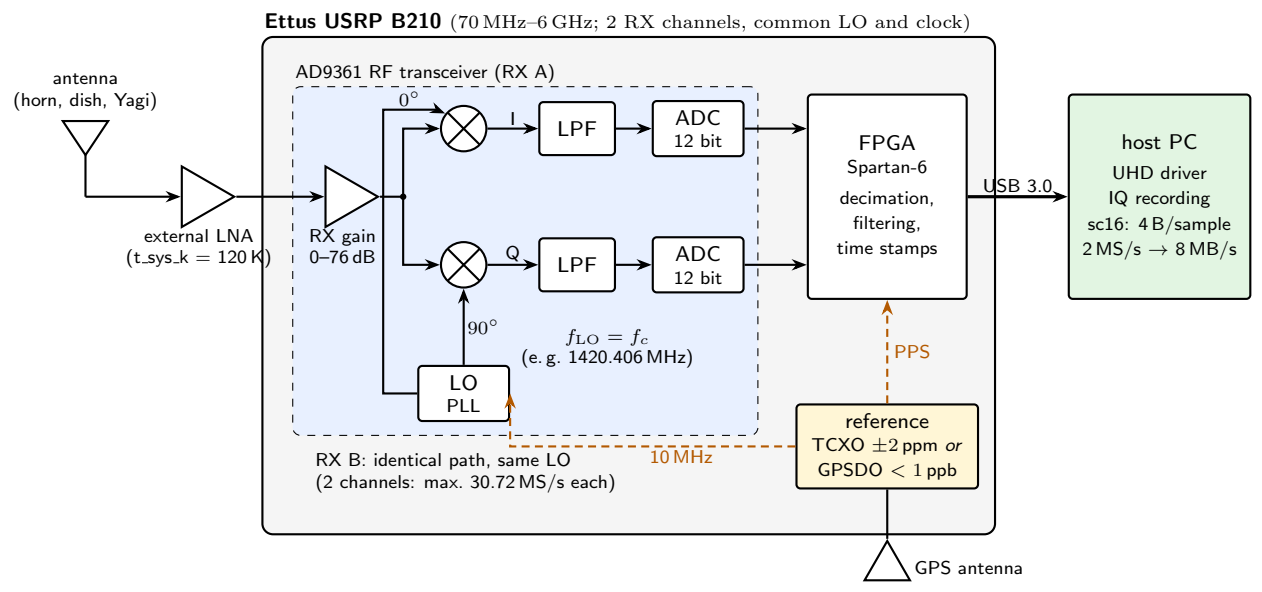

Ettus USRP B210 (70 MHz–6 GHz; 2 RX channels, common LO and clock)
antenna
(horn, dish, Yagi)
external LNA
(t_sys_k = 120 K)
AD9361 RF transceiver (RX A)
RX gain
0–76 dB
0°
I
LPF
ADC
12 bit
Q
LPF
ADC
12 bit
90°
LO
PLL
$f_{LO} = f_c$
(e. g. 1420.406 MHz)
RX B: identical path, same LO
(2 channels: max. 30.72 MS/s each)
FPGA
Spartan-6
decimation,
filtering,
time stamps
USB 3.0
host PC
UHD driver
IQ recording
sc16: 4 B/sample
2 MS/s → 8 MB/s
PPS
reference
TCXO ±2 ppm or
GPSDO < 1 ppb
10 MHz
GPS antenna